Jiahao Yu

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Education

Northwestern University — **Evanston, United States**
Ph.D., Computer Science — *Jan. 2022 – Now*

- Advisor: Prof. Xinyu Xing
- GPA: 3.85/4.0

Shanghai Jiao Tong University — **Shanghai, China**
B.S, School of Electronic Information and Electrical Engineering — *Sep. 2017 – July 2021*

- Advisor: Prof. Liyao Xiang
- Zhiyuan Honor Program
- GPA: 3.6/4.0

Professional Appointments

New York University Abu Dhabi — **Abu Dhabi, United Arab Emirates**
Tenure-Track Assistant Professor (TTAP), Computer Engineering — *Aug. 1, 2026 – Present*

- Core Faculty, Center for Cyber Security Abu Dhabi
- Lead Principal Investigator (PI), Secure Reasoning Lab

Research Interests

My research is situated at the intersection of AI and security, with a broad focus on leveraging artificial intelligence to advance cybersecurity capabilities. Specifically, I concentrate on using Large Language Models (LLMs) and Reinforcement Learning (RL) for critical security tasks, including automated program patch generation, vulnerability identification, and securing blockchain systems. From a machine learning perspective, my work aims to improve LLMs' explainability, adversarial robustness, noise resistance, and overall effectiveness of these models and algorithms.

Published Work

*Denotes equal contribution.

2026

Decoupled Alignment for Robust Plug-and-Play Adaptation
- Haozheng Luo*, **Jiahao Yu***, Wenxin Zhang, Jialong Li, Chenghao Qiu, Yimin Wang, Eric Hanchen Jiang, Jerry Yao-Chieh Hu, Yan Chen, Binghui Wang, Xinyu Xing, Han Liu
- *In Proceedings of the Conference on Language Modeling (COLM) 2026*

Contrastive Reasoning Alignment: Reinforcement Learning from Hidden Representations
- Haozheng Luo, Yimin Wang, **Jiahao Yu**, Binhui Wang, Yan Chen
- *In Proceedings of the International Conference on Machine Learning (ICML) 2026*

Locus: Agentic Predicate Reasoning for Directed Fuzzing
- Jie Zhu, Chihao Shen, Ziyang Li, **Jiahao Yu**, Yizheng Chen, Kexin Pei
- *In Proceedings of the International Conference on Software Engineering (ICSE) 2026*

PROMPTFUZZ: Harnessing Fuzzing Techniques for Robust Testing of Prompt Injection in LLMs
- **Jiahao Yu***, Yangguang Shao*, Hanwen Miao, Junzheng Shi
- *Transactions on Information Forensics & Security*

2025

Mind the Inconspicuous: Revealing the Hidden Weakness in Aligned LLMs' Ethical Boundaries
- **Jiahao Yu**, Haozheng Luo, Jerry Yao-Chieh Hu, Wenbo Guo, Yan Chen, Han Liu, Xinyu Xing
- *In Proceedings of the USENIX Security Symposium (USENIX Security) 2025 (Long Talk)*

PatchAgent: A Practical Program Repair Agent Mimicking Human Expertise
- Zheng Yu, Ziyi Guo, Yuhang Wu, **Jiahao Yu**, Meng Xu, Dongliang Mu, Yan Chen, Xinyu Xing
- *In Proceedings of the USENIX Security Symposium (USENIX Security) 2025 (Long Talk)*
- This work was the foundation of the auto patch generation for our winning solution (Team 42-b3yond-bug) in the DARPA AI Cyber Challenge (AIxCC), securing a *$1,000,000 grant for proposal* and *$2,000,000 prize in semi-final*.
- This work was selected as Runner-up in Technical Impact in CSAW 2025.

The Illusion of Role Separation: Hidden Shortcuts in LLM Role Learning (and How to Fix Them)
- Zihao Wang, Yibo Jiang, **Jiahao Yu**, Heqing Huang
- *In Proceedings of the International Conference on Machine Learning (ICML) 2025*

GPO: Learning from Critical Steps to Improve LLM Reasoning
- **Jiahao Yu**, Zelei Cheng, Xian Wu, Xinyu Xing
- *In Proceedings of the Conference on Neural Information Processing Systems (NeurIPS) 2025*
- This work was featured in *MIT Technology Review China a* and *MIT Technology Review China b* (October 2025).

BlockScan: Detecting Anomalies in Blockchain Transactions
- **Jiahao Yu**, Xian Wu, Hao Liu, Wenbo Guo, Xinyu Xing
- *In Proceedings of the Conference on Neural Information Processing Systems (NeurIPS) 2025*

Knowledge-Distilled Memory Editing for Plug-and-Play LLM Alignment
- Haozheng Luo*, **Jiahao Yu***, Wenxin Zhang*, Jialong Li, Jerry Yao-Chieh Hu, Yan Chen, Binhui Wang, Xinyu Xing, Han Liu
- *ICML 2025 Workshop on the Impact of Memorization on Trustworthy Foundation Models (ICLM@MemFM)*

UTF: Undertrained Tokens as Fingerprints A Novel Approach to LLM Identification
- Jiacheng Cai*, **Jiahao Yu***, Yangguang Shao, Yuhang Wu, Xinyu Xing
- *ACL 2025 Workshop on LLMSec (ACL@LLMSec)*

2024

Soft-Label Integration for Robust Toxicity Classification
- Zelei Cheng*, Xian Wu*, **Jiahao Yu***, Shuo Han, Xin-Qiang Cai, Xinyu Xing
- *In Proceedings of the Conference on Neural Information Processing Systems (NeurIPS) 2024*
- This work was featured in *MIT Technology Review China* (December 2023).

LLM-Fuzzer: Scaling Assessment of Large Language Model Jailbreaks
- **Jiahao Yu**, Xingwei Lin, Zheng Yu, Xinyu Xing
- *In Proceedings of the USENIX Security Symposium (USENIX Security) 2024*
- Originally released as GPTFuzzer, this work received the *Geekcon 2023 Annual Themed Debate Breakthrough Award*.
- The tool has been widely adopted by major LLM providers including OpenAI, ByteDance, Ant Group, ScaleAI and Anthropic, and is integrated into Microsoft Azure's PyRIT.
- The associated open-source tool has been downloaded over 146,000 times.

RICE: Breaking Through the Training Bottlenecks of Reinforcement Learning with Explanation
- Zelei Cheng*, Xian Wu*, **Jiahao Yu***, Sabrina Yang, Gang Wang, Xinyu Xing
- *In Proceedings of the International Conference on Machine Learning (ICML Spotlight) 2024*

BandFuzz: A Practical Framework for Collaborative Fuzzing with Reinforcement Learning
- Wenxuan Shi, Hongwei Li, **Jiahao Yu**, and Wenbo Guo, Xinyu Xing
- *ICSE Workshop on Search-Based and Fuzz Testing (ICSE@SBFT) 2024*
- This work won the **First Prize at the SBFT 2024 Fuzzing Competition**, ranking 1st across all evaluation metrics.
- It was featured in *McCormick School of Engineering News*.
- It served as the foundational fuzzing component for our winning solution in the DARPA AI Cyber Challenge (AIxCC) semi-final.
- In the AIxCC final, this framework discovered *the highest number of zero-day vulnerabilities among all competitors*.

Assessing Prompt Injection Risks in 200+ Custom GPTs
- **Jiahao Yu**, Yuhang Wu, Dong Shu, Mingyu Jin, Xinyu Xing
- *ICLR 2024 Workshop on Secure and Trustworthy Large Language Models* (*ICLR@SeTLLM*)
- This research was featured in *WIRED* (November 2023).

2023

StateMask: Explaining Deep Reinforcement Learning through State Mask
- Zelei Cheng*, Xian Wu*, **Jiahao Yu***, Wenbo Guo, Wenhai Sun, Xinyu Xing
- *In Proceedings of the Conference on Neural Information Processing Systems* (*NeurIPS*) *2023*

AIRS: Explanation for Deep Reinforcement Learning based Security Applications
- **Jiahao Yu**, Wenbo Guo, Qi Qin, Gang Wang, Ting Wang, Xinyu Xing
- *In Proceedings of the USENIX Security Symposium* (*USENIX Security*) *2023*

Publications Prior to Ph.D. Program

2021

Matrix Gaussian Mechanism for Differentially-Private Learning
- Jungang Yang, Liyao Xiang, **Jiahao Yu**, Xinbing Wang, Bin Guo, Bin Guo, Zhetao Li, Baochun Li
- *In IEEE Transactions on Mobile Computing* (*TMC*) *2021*

Speedup robust graph structure learning with low-rank information
- Hui Xu, Liyao Xiang, **Jiahao Yu**, Anqi Cao, Xinbing Wang
- *In Proceedings of the 30th ACM International Conference on Information & Knowledge Management* (*CIKM*) *2021*

2020

Voiceprint Mimicry Attack Towards Speaker Verification System in Smart Home
- Lei Zhang, Yan Meng, **Jiahao Yu**, Chong Xiang, Brandon Folk, Haojin Zhu
- *In Proceedings of the IEEE International Conference on Computer Communications* (*INFOCOM*) *2020*

Research on Application of Artificial Intelligence Technology in Electrical Automation Control
- Chao Jiang, Xiaorui Xiong, Tanqing Zhu, Jiajia Cao, **Jiahao Yu**
- *Journal of Physics: Conference Series, Vol. 1578, 2020*

Unpublished Manuscripts & Preprints

2026

Building Coding Agents via Entropy-Enhanced Multi-Turn Preference Optimization
- **Jiahao Yu**, Zelei Cheng, Xian Wu, Xinyu Xing
- *arXiv preprint arXiv:2509.12434*
- It ranks *1st in SWEBench-Lite* and *5th in SWEBench-Verified*(only suppressed by models 10x larger than us) official open-weight leaderboard.

2025

A Survey on Explainable Deep Reinforcement Learning
- Zelei Cheng*, **Jiahao Yu***, Xinyu Xing
- *arXiv preprint arXiv:2502.06869*

BandFuzz: An ML-powered Collaborative Fuzzing Framework
- Wenxuan Shi, Hongwei Li, **Jiahao Yu**, Xinqian Sun, Wenbo Guo, Xinyu Xing
- *arXiv preprint arXiv:2507.10845*

GenoArmory: A Unified Evaluation Framework for Adversarial Attacks on Genomic Foundation Models
- Haozheng Luo, Chenghao Qiu, Yimin Wang, Shang Wu, **Jiahao Yu**, Han Liu, Binghui Wang, Yan Chen
- *https://github.com/MAGICS-LAB/GenoArmory*

POISONCRAFT: Practical Poisoning of Retrieval-Augmented Generation for Large Language Models
- Yangguang Shao, Xinjie Lin, Haozheng Luo, Chengshang Hou, Gang Xiong, **Jiahao Yu**, Junzheng Shi
- *arXiv preprint arXiv:2505.06579*

2023

GPTFUZZER : Red Teaming Large Language Models with Auto-Generated Jailbreak Prompts
- **Jiahao Yu**, Xingwei Lin, Xinyu Xing
- *arXiv preprint arXiv:2309.10253*

Professional Service

- **Workshop Organizer**
  - The 2nd Workshop on Efficient Reasoning at COLM 2026
- **Program Committee Member**
  - USENIX Security Symposium ('27 Cycle 1)
  - USENIX Security Symposium ('26 Cycle 1)
  - USENIX Security Symposium ('26 Cycle 2)
  - Conference on Computer and Communications Security (CCS) 2026 (Track A)
  - Conference on Computer and Communications Security (CCS) 2026 (Track B)
  - LLM4Code@ICSE 2027
  - LLM4Code@ICSE 2026
- **Area Chair**
  - ES-Reasoning@ICLR 2026
- **Reviewer**
  - Conference on Language Modeling (COLM) 2026
  - AAAI Conference on Artificial Intelligence (AAAI) 2026
  - ACM Computing Surveys (CSUR) 2026
  - IEEE Transactions on Dependable and Secure Computing (TDSC) 2026
  - IEEE Transactions on Software Engineering (TSE) 2026
  - International Conference on Learning Representations (ICLR) 2025 (**Notable Reviewer**), 2026
  - IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2026, 2027
  - Conference on Neural Information Processing Systems (NeurIPS) 2024, 2025
  - International Conference on Machine Learning (ICML) 2025
  - International Conference on Artificial Intelligence and Statistics (AISTATS) 2025, 2026
  - The Web Conference (WWW) 2025
  - ICLR Workshop on Secure and Trustworthy Large Language Models 2024
  - IEEE Transactions on Information Forensics and Security (TIFS) 2025
  - IEEE Transactions on Knowledge and Data Engineering (TKDE) 2024
  - Journal of Orthopaedic Surgery 2025
- **Artifact Evaluation Committee Member**
  - USENIX Security Symposium 2025 (Cycle 1, Cycle 2), 2026 (Cycle 1, Cycle 2)
  - Network and Distributed System Security Symposium (NDSS) 2025 (Summer, Fall)
- **External Reviewer**
  - ACM Conference on Computer and Communications Security (CCS) 2023 (Track B), 2022 (Track B)
  - IEEE Symposium on Security and Privacy (S&P) 2025 (Cycle 2), 2023 (Cycle 3), 2022 (Poster)
  - International Conference on Automated Software Engineering (ASE) 2024
  - International Conference on Learning Representations (ICLR) 2024
  - International Symposium on Research in Attacks, Intrusions and Defenses (RAID) 2022
  - Network and Distributed System Security Symposium (NDSS) 2025
  - USENIX Workshop on Offensive Technologies (WOOT) 2024

Awards

- ICLR Notable Reviewer 2025
- CSAW Runner-up in Technical Impact 2025
- DARPA AIxCC Semi-Finalist (Team 42-b3yond-bug) 2024
- OpenAI Researcher Access Program 2024
- First Prize, SBFT Fuzzing Competition 2024
- Geekcon Breakthrough Award 2023

Working Experiences

LLM Red-teaming Benchmarking **San Francisco, USA**
ByteDance *Sep. 2024 – Dec. 2024*

- Research Intern
- Advisor: Dr. Zhenqing Luo

LLM for Vulnerability Detection **Chicago, USA**
University of Chicago *June. 2024 – Sep. 2024*

- Research Assistant
- Advisor: Dr. Kexin Pei

Automatically Red-teaming Large Language Models **Hangzhou, China**
Ant Group *June. 2023 – Aug. 2023*

- Research Intern
- Advisor: Xingwei Lin

Adversarial Attack against PowerShell Malware Detector **Beijing, China**
MSRA *Aug. 2020 – Mar. 2021*

- Research Intern
- Advisor: Dr. Bin Zhu